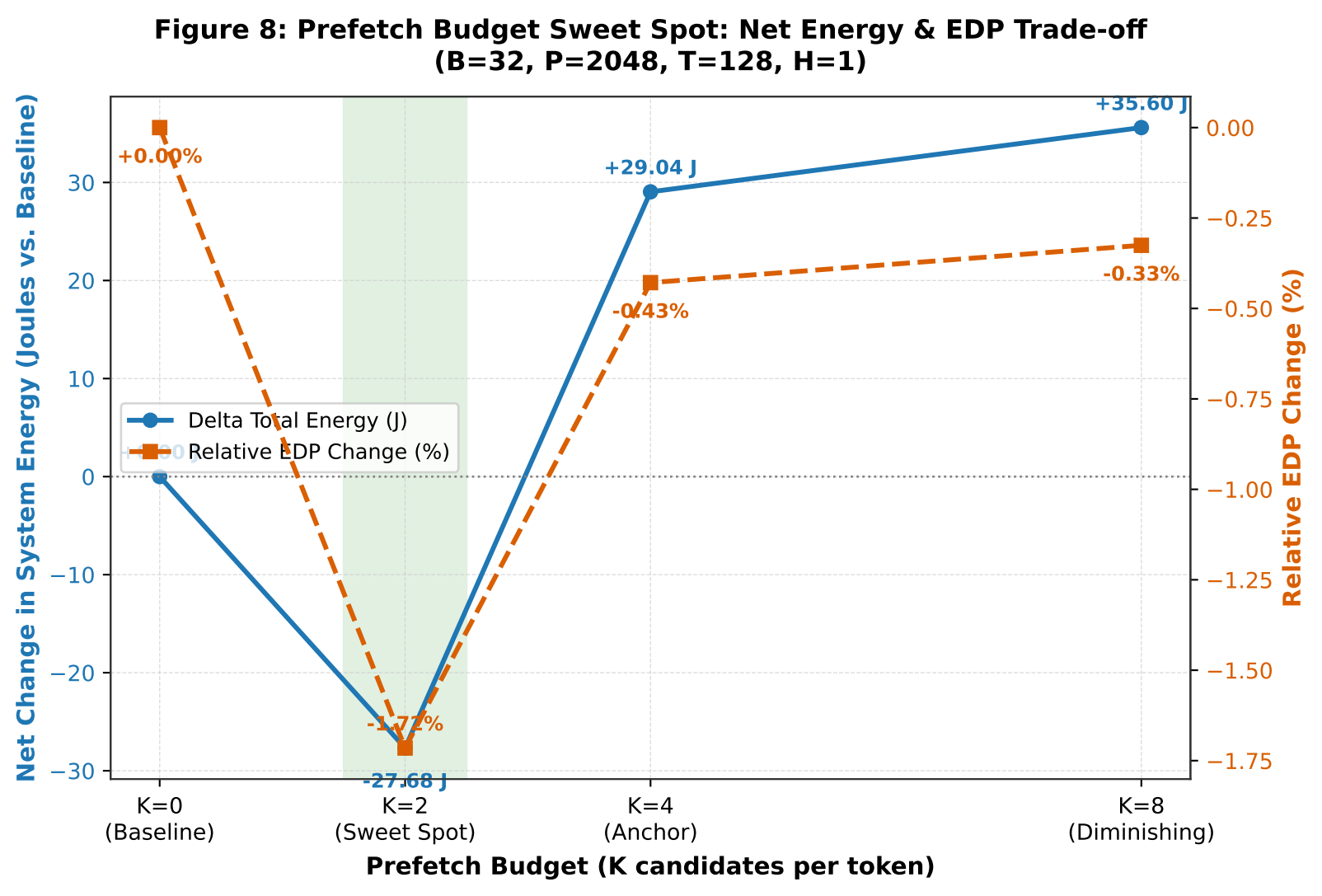
Figure 8: Prefetch Budget Sweet Spot: Net Energy & EDP Trade-off
(B=32, P=2048, T=128, H=1)
Net Change in System Energy (Joules vs. Baseline)
Relative EDP Change (%)
Prefetch Budget (K candidates per token)
Delta Total Energy (J)
Relative EDP Change (%)
+0.00%
-27.68 J
-1.72%
+29.04 J
-0.43%
+35.60 J
-0.33%
K=0
(Baseline)
K=2
(Sweet Spot)
K=4
(Anchor)
K=8
(Diminishing)
30
20
10
0
−10
−20
−30
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75